

Bank Loan Analytics Dashboard – Business Analyst Project Report

Project Objective

To analyze a bank's historical loan portfolio and develop an interactive dashboard to help management monitor loan performance, understand borrower characteristics, identify lending trends, and support data-driven business decisions.

Business Problem Statement

Banks process thousands of loan applications and generate large volumes of lending data. Without a centralized reporting system, it becomes difficult to monitor loan performance, evaluate borrower risk, and identify lending trends. This project addresses that challenge by transforming raw loan data into an interactive dashboard that provides meaningful insights for business users.

Dataset Overview

The dataset contains approximately 38,000 loan applications with fields such as loan amount, funded amount, total payment, loan status, borrower grade, annual income, interest rate, debt-to-income ratio (DTI), home ownership, verification status, loan purpose, issue date, and state. These attributes help evaluate loan performance, borrower behaviour, and portfolio quality.

Tools Used and My Contribution

Microsoft Excel was used to understand the dataset and prepare a data dictionary. Power Query was used for data cleaning, checking missing values, duplicates, validating data types, and transforming the dataset. Power BI was used to build the interactive dashboard, while DAX was used to create business KPIs and calculations.

Project Workflow

The workflow followed was: Data Understanding → Data Dictionary Creation → Data Cleaning using Power Query → Data Validation → Import into Power BI → DAX Measure Creation → Dashboard Design → Business Insight Generation.

Key Performance Indicators (KPIs)

The dashboard includes Total Loan Applications, Total Funded Amount, Total Amount Received, Average Interest Rate, and Average Debt-to-Income (DTI). These KPIs provide a high-level view of lending performance and portfolio quality.

Business Questions Answered

The dashboard answers questions such as: How many loan applications were processed? Which borrower grades received the highest funding? What are the most common loan purposes? Which states have the highest loan amounts? What is the distribution of loan status? How do home ownership and verification status relate to the borrower portfolio?

Key Business Insights

The analysis provides a centralized view of the loan portfolio, enabling stakeholders to identify high-performing borrower segments, monitor loan status distribution, understand customer borrowing behaviour, evaluate regional lending patterns, and track important financial KPIs. These insights support better operational monitoring and strategic decision-making.

Key Findings:

- Total Loan Applications exceeded 38,000.
- Fully Paid loans represented the largest share of the portfolio.
- Grade B borrowers received the highest funded amount.
- Debt Consolidation was the most common loan purpose.
- California contributed the highest funded amount.
- Average DTI remained within an acceptable range.

Business Recommendations

1. Loan Status Analysis

Observation

The dashboard shows that Fully Paid loans make up the largest portion of the portfolio, while Charged-Off loans represent the smallest but highest-risk segment.

Recommendation

Continue the current lending strategy for customer segments with strong repayment history. At the same time, create an early-warning monitoring process for Charged-Off loans to identify high-risk borrowers before default.

Business Impact

Improves portfolio quality and reduces future credit losses.

2. Borrower Grade

Observation

Grade B borrowers received the highest funded amount in the portfolio, indicating a strong balance between lending demand and acceptable credit quality.

Recommendation

Increase lending opportunities for consistently high-performing grades while maintaining stricter eligibility checks for lower grades.

Business Impact

Maximizes loan growth while balancing credit risk.

3. Loan Purpose

Observation

The dashboard highlights the most common loan purpose.

Recommendation

Debt Consolidation represents the largest borrowing purpose. The bank can introduce customized refinancing products and promotional offers for this segment and promotional campaigns for the highest-demand loan purpose while reviewing pricing for less popular products.

Business Impact

Improves customer acquisition and increases loan disbursement.

4. State-wise Analysis

Observation

California contributed the highest funded amount. The bank can prioritize marketing campaigns and branch expansion in California while identifying similar markets for future growth.

Recommendation

Expand branch operations and targeted marketing in high-performing states while identifying similar markets with growth potential.

Business Impact

Supports business expansion and increases lending revenue.

5. Home Ownership

Observation

RENT customers account for the largest borrower segment. Introducing flexible repayment plans designed for renters could improve customer acquisition and retention.

Recommendation

Design loan products and repayment plans that better match the financial needs of this customer segment.

Business Impact

Improves customer satisfaction and increases approval opportunities.

6. Verification Status

Observation

The dashboard compares verified and non-verified borrowers.

Recommendation

Since Verified borrowers represent the majority of approved loans, maintaining strict verification policies while simplifying verification for low-risk applicants can improve operational efficiency.

Business Impact

Reduces fraud risk without unnecessarily delaying approvals.

7. Interest Rate & DTI

Observation

The dashboard tracks Average Interest Rate and Average Debt-to-Income Ratio.

Recommendation

Use DTI thresholds along with credit grades when evaluating new applications to ensure borrowers have sufficient repayment capacity.

Business Impact

Improves lending decisions and reduces default probability.

Business Outcome

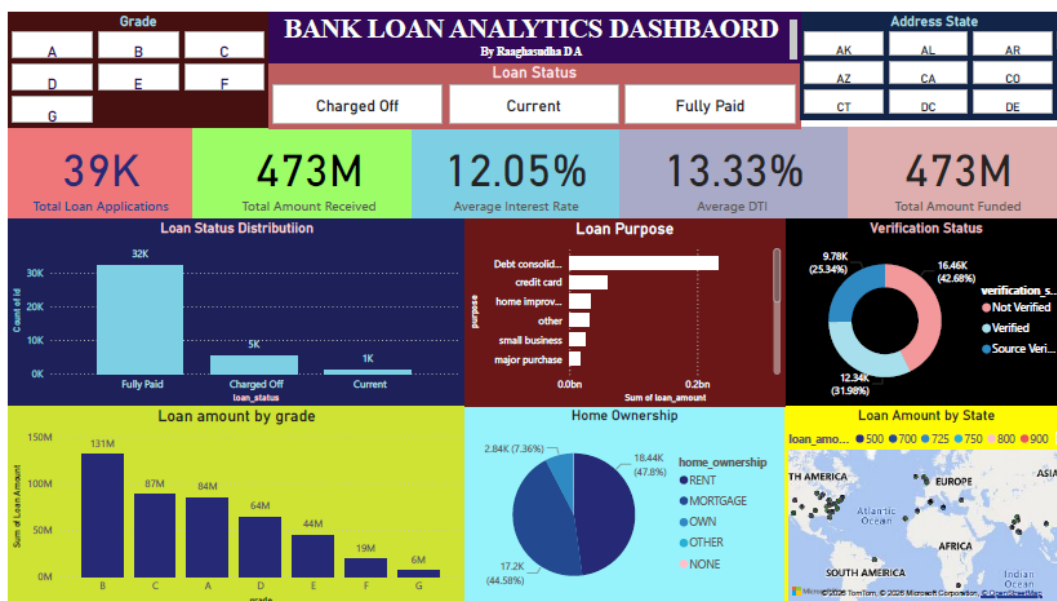
The final deliverable was an interactive Power BI dashboard that converts raw banking data into actionable insights. It enables management to monitor portfolio performance, identify lending trends, evaluate borrower segments, support informed lending decisions, and improve credit risk management.

Skills Demonstrated

Business Analysis, Data Cleaning, Data Transformation, Data Visualization, Dashboard Design, KPI Development, DAX, Power Query, Microsoft Excel, and Power BI.

Conclusion

This project demonstrates the complete lifecycle of a business analytics solution, from understanding business requirements and preparing data to creating dashboards and communicating business insights. It strengthened my analytical thinking, problem-solving skills, and ability to support data-driven decision-making.



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